

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
```

```
df = sns.load_dataset('iris')
print("First five rows of the dataset:")
print(df.head())
```

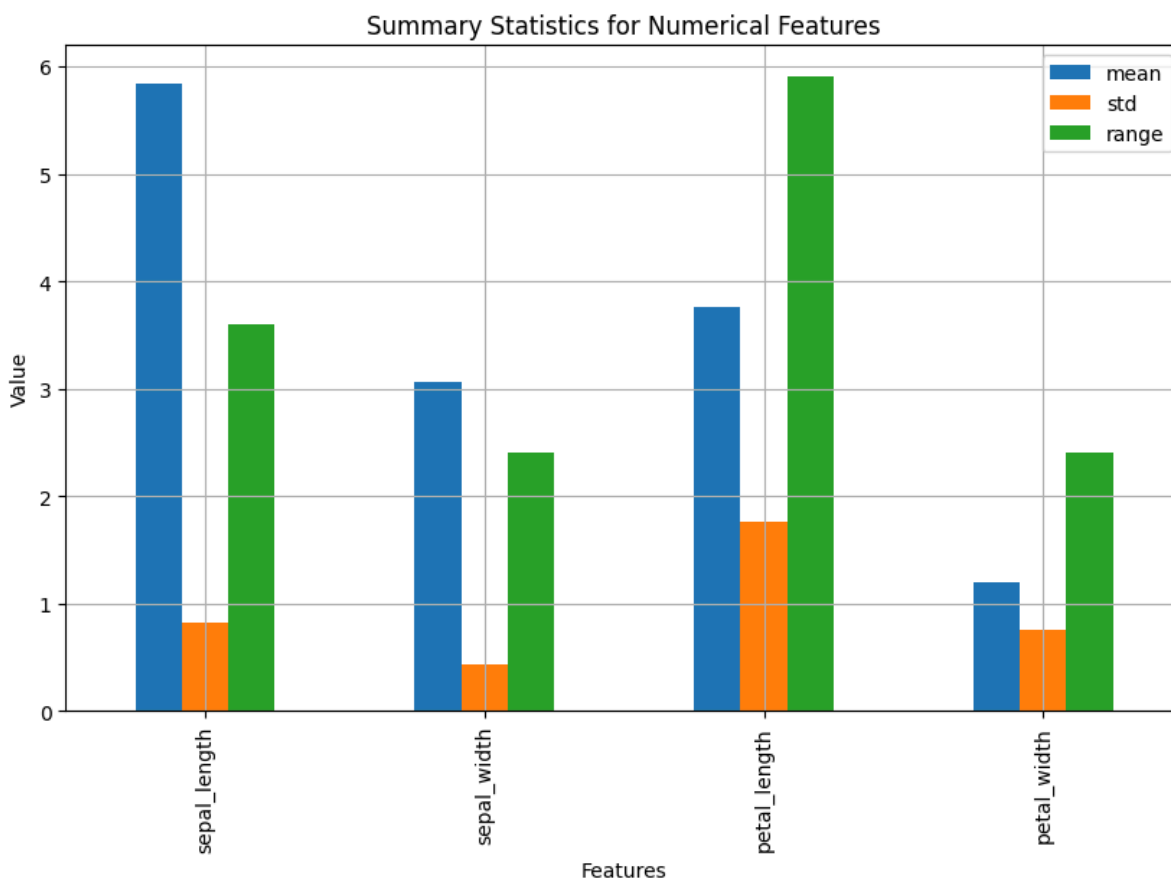
```
First five rows of the dataset:
  sepal_length  sepal_width  petal_length  petal_width  species
0          5.1           3.5           1.4           0.2   setosa
1          4.9           3.0           1.4           0.2   setosa
2          4.7           3.2           1.3           0.2   setosa
3          4.6           3.1           1.5           0.2   setosa
4          5.0           3.6           1.4           0.2   setosa
```

```
summary_stats = df.describe().T
summary_stats['range'] = summary_stats['max'] - summary_stats['min']
summary_stats['variance'] = df.select_dtypes(include=np.number).var()
summary_stats['25%'] = df.select_dtypes(include=np.number).quantile(0.25)
summary_stats['75%'] = df.select_dtypes(include=np.number).quantile(0.75)
print("\nSummary Statistics:")
print(summary_stats)
```

```
Summary Statistics:
count      mean      std  min  25%  50%  75%  max  range \
sepal_length  150.0  5.843333  0.828066  4.3  5.1  5.80  6.4  7.9   3.6
sepal_width   150.0  3.057333  0.435866  2.0  2.8  3.00  3.3  4.4   2.4
petal_length  150.0  3.758000  1.765298  1.0  1.6  4.35  5.1  6.9   5.9
petal_width   150.0  1.199333  0.762238  0.1  0.3  1.30  1.8  2.5   2.4

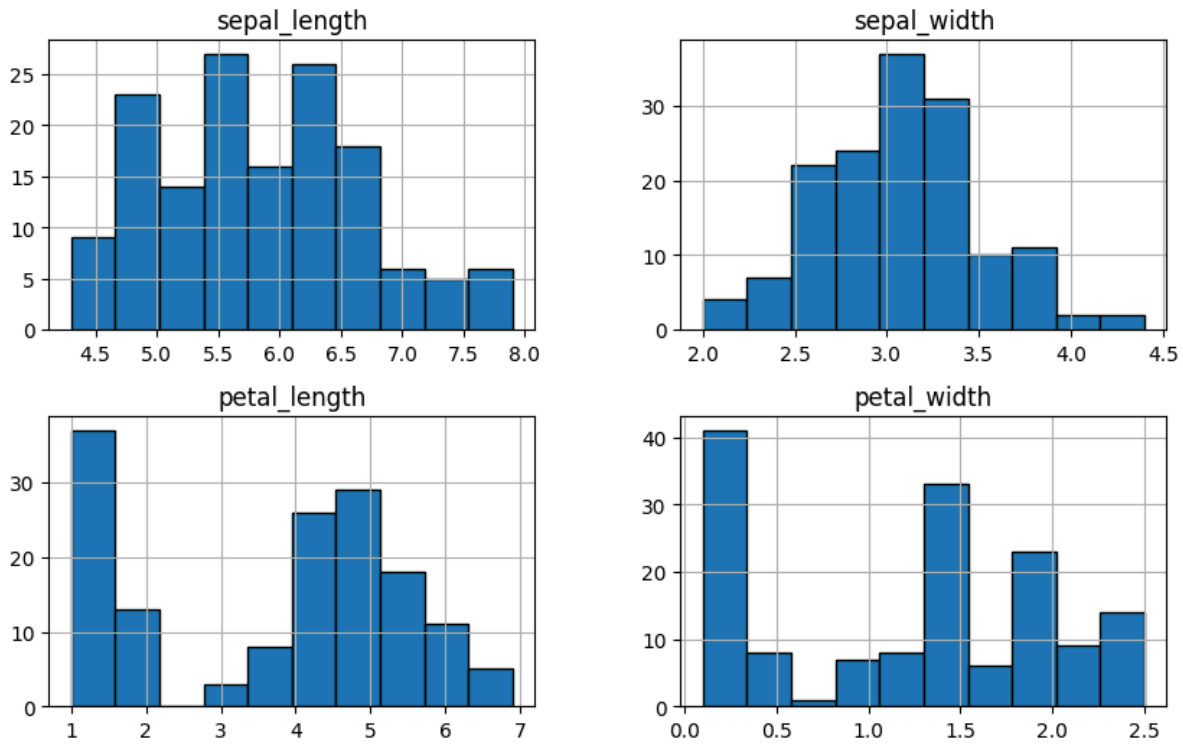
variance
sepal_length  0.685694
sepal_width   0.189979
petal_length  3.116278
petal_width   0.581006
```

```
summary_stats[['mean', 'std', 'range']].plot(kind='bar', figsize=(10,6))
plt.title('Summary Statistics for Numerical Features')
plt.ylabel('Value')
plt.xlabel('Features')
plt.grid(True)
plt.show()
```



```
df.hist(figsize=(10,6),bins=10, edgecolor='black')
plt.suptitle('Histograms of Each Feature', fontsize=14)
plt.show()
```

Histograms of Each Feature



```
plt.figure(figsize=(8,6))
sns.boxplot(data=df.drop(columns=['species']), palette='Set2')
plt.title('Boxplots of Numerical Features')
plt.show()
```

Boxplots of Numerical Features

